nature computational science

Article

https://doi.org/10.1038/s43588-026-01029-2

Structure-alignment-driven cross-graph modeling for functional RNA design

Received: 8 August 2025

Accepted: 30 June 2026

Published online: 30 July 2026

Check for updates

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RNAs are critical for biological processes, and their functions are closely tied to their three-dimensional structures. The design of RNA sequences that fold into target three-dimensional structures is a complex challenge due to the dynamic and unstable nature of RNA structures. Here we present an RNA design approach, AlignIF, that leverages multiple structure alignment with cross-graph modeling and carefully hand-designed features to capture evolutionarily conserved structural patterns at the structural level, facilitating RNA sequence design. AlignIF outperforms existing state-of-the-art methods in test sets. Notably, it enables the design of entire RNA families rather than being restricted to recapitulating native sequences. Furthermore, AlignIF successfully designs functional RNA fluorescent aptamers and self-cleaving ribozymes that exhibit measurable activity without requiring any postdesign filtering. For Mango-I, two of ten designed aptamers showed enhanced fluorescence, and two others exhibited improved binding affinity compared with the wild type. For the more challenging *i*Mango-III, all designs were fluorescent, including one with a sequence identity of ~30% relative to the wild type. The results highlight the potential of AlignIF for functional RNA engineering.

RNAs play critical roles in cellular functions as intermediates in protein synthesis[1], regulators of gene expression[2], molecular sensors[3] and catalysts that cleave nucleotide chains[4]. These diverse roles require RNAs to fold into specific three-dimensional (3D) structures. Deciphering these complex 3D conformations has emerged as a central challenge in RNA biology, offering critical insights into their mechanisms and potential applications in RNA therapeutics.

Recently, RNA sequence design under structural constraints has emerged as a powerful computational approach for addressing this challenge, including RDesign[5], RiboDiffusion[6], gRNAde[7] and RhoDesign[8]. Enabling the generation of diverse RNA sequences that fold into a predetermined 3D structure offers a transformative framework for dissecting RNA sequence–structure–function relationships. This capability enables a systematic interrogation of sequence features, identifying those essential for structural integrity as well as those that can tolerate variation without compromising biological functions. As a result, structure-based RNA sequence design not only facilitates the rational engineering of functional RNAs but also advances our understanding of how RNA sequence encodes biological functions.

However, RNA conformations are inherently dynamic[9], flexible[10] and unstable[11], which restricts the availability of experimentally determined 3D structural data[12]. This scarcity creates a substantial gap

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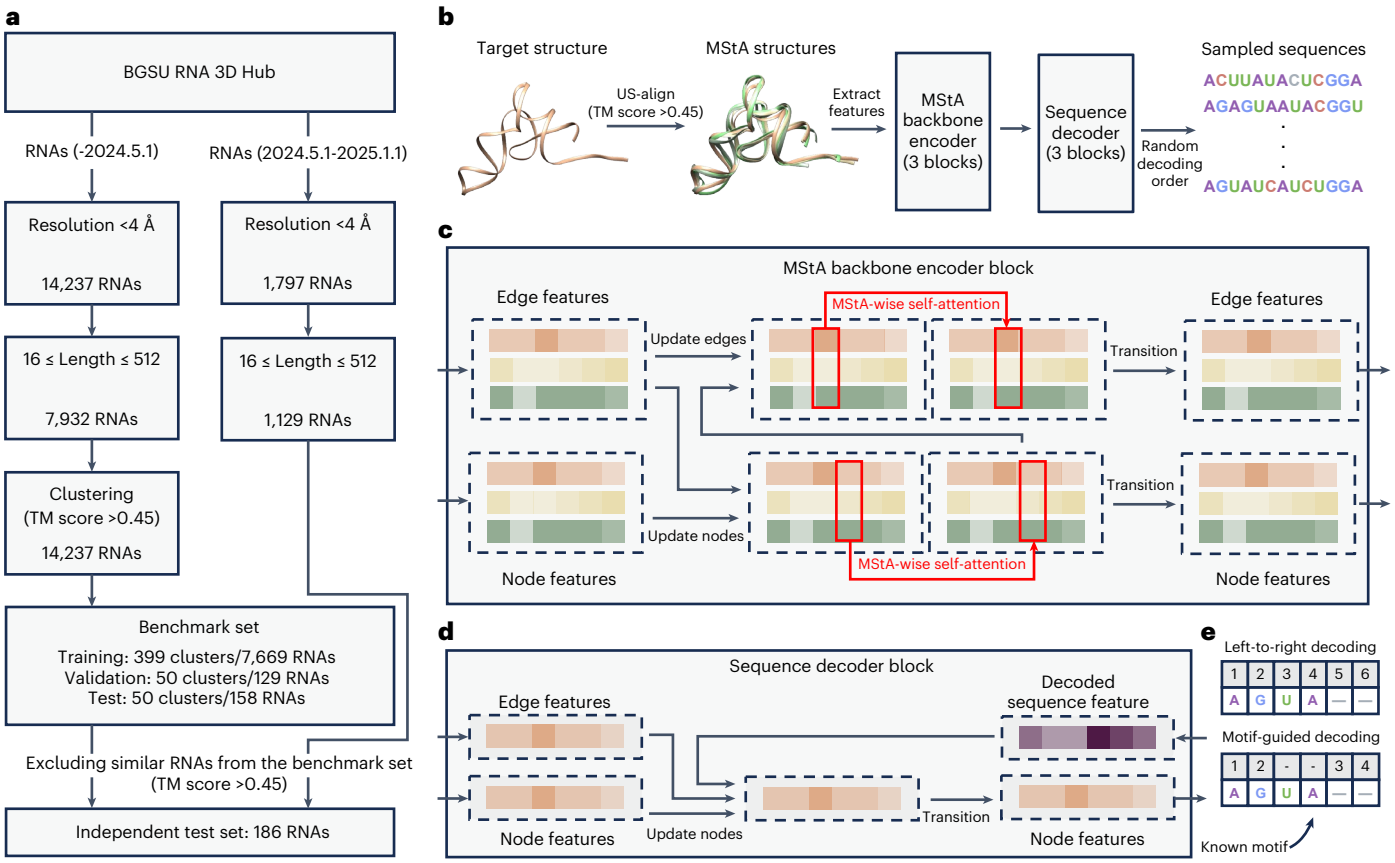

Fig. 1 | Overview of AlignIF architecture. a, Construction of the benchmark set and independent test set from BGSU RNA 3D Hub. **b**, MStA structures are retrieved from the dataset, aligned to the target structure using US-align, and processed by the MStA backbone encoder and sequence decoder to generate sequences. **c**, The encoder employs message-passing and MStA-wise self-attention to aggregate structural information, with distinct colors indicating features from different MStA structures. **d**, The random-ordered autoregressive decoder uses node and edge features from the target structure to iteratively generate nucleotides in a flexible, randomized order. **e**, AlignIF supports two sampling strategies: (1) conventional left-to-right decoding and (2) motif-guided decoding, which integrates motif information into the generation of nucleotides when available.

between the known sequence–structure relationships and their true distribution, often resulting in severe overfitting and limiting the generalizability to novel or unseen RNA molecules. Existing design approaches primarily focus on a single target RNA structure, neglecting potentially informative similarities across related RNA structures. Although gRNAde (multistate) incorporates multiple conformational states of the same RNA, it faces critical limitations. In most cases, only a single conformational state is accessible, which limits the general applicability. Thus, effectively leveraging the limited structural data to improve RNA sequence design remains a key unresolved challenge. Furthermore, existing design approaches are still not fully experimentally verified for a full range of functional RNAs, which are limited to RNA fluorescent aptamers[8]. In addition, the sugar ring in RNA structures is inherently flexible, which can be applied to facilitate the structure-based RNA sequence design.

Structure prediction typically leverages multiple sequence alignments (MSAs) derived from the target sequence to capture evolutionary information, thereby improving the accuracy for both proteins and RNAs[13–15]. For structure-based RNA sequence design, the inverse problem of RNA structure prediction, we adapt this principle to learn evolutionary patterns at the structural level. Here, we present AlignIF, an RNA design framework that leverages multiple structure alignment (MStA) and cross-graph modeling to generate sequences. AlignIF combines a graph neural network (GNN)-based MStA backbone encoder equipped with a cross-graph attention mechanism, hand-designed geometric features from aligned structures, and a random-ordered autoregressive decoder. This architecture captures evolutionarily conserved structural patterns and enables effective RNA sequence design. We experimentally verified 23 designed RNA fluorescent aptamers and 7 self-cleaving ribozymes, all showing expected activity, demonstrating strong in silico and experimental performance.

Results

Overview of AlignIF for functional RNA design

AlignIF advances structure-based RNA sequence design by integrating MStA data, and it designs an MStA backbone encoder to capture structural evolutionary conservation patterns. To objectively evaluate AlignIF, we curated a benchmark dataset from the Protein Data Bank (PDB) using the BGSU RNA 3D Hub[16], selecting structures with resolutions better than 4 Å and lengths between 16 and 512 nucleotides. This dataset, comprising 7,932 RNAs deposited before 1 May 2024, was clustered using US-align[17] at a template modeling score (TM score) threshold of 0.45, yielding 499 nonredundant clusters, which are split into a benchmark train set, a benchmark validation set and a benchmark test set. For independent evaluation, we assembled an independent test set of 186 RNAs deposited between 1 May 2024 and 1 January 2025, ensuring the structural dissimilarity (TM score <0.45) from the benchmark set (Fig. 1a). To further evaluate AlignIF for in silico predicted RNA structures, we evaluate it on 50 RNA structures predicted by AlphaFold3 from RNA sequences ranging from 10 to 510 nucleotides.

The AlignIF pipeline processes RNA structures as follows (Fig. 1b). Given a target RNA structure, we run US-align to obtain its MStA (TM score ≥0.45), which undergoes feature extraction to derive node and edge features, including direction, orientation, dihedral angles and distances (see '3D RNA structure representation' section in the Methods). These features are encoded using a GNN-based backbone encoder, which iteratively refines them through message-passing and MStA-wise self-attention across multiple geometric graphs derived from the multiple aligned structures (Fig. 1c). A decoder then autoregressively generates nucleotides in randomized order, conditioned on the target's structural features (Fig. 1d), inspired by ProteinMPNN[18]. This random decoding strategy reduces overfitting and supports multiple decoding order strategies, allowing AlignIF to prioritize known motifs before generating remaining nucleotides conditioned on these motifs (Fig. 1e).

Recovering native RNA sequences

To demonstrate the effectiveness of AlignIF, we compared AlignIF against several state-of-the-art structure-based RNA sequence design approaches, including RDesign[5], RiboDiffusion[6], gRNAde (single-state and multistate versions)[7] and RhoDesign[8] on the benchmark and independent test set. In addition, we evaluated AlignIF in the absence of any MStA, representing a realistic worst-case scenario for designing sequences of highly novel RNA folds with no homologous structures available. For fair comparison, we retrained these baseline models on the same training dataset as AlignIF. Performance was evaluated using two key metrics: sequence recovery rate and perplexity. The recovery rate measures the model's ability to design sequences that match the native sequence, and the perplexity quantifies the model's uncertainty in predicting the native sequence, derived from the probability distribution of predicted nucleotides. Notably, perplexity cannot be calculated for RiboDiffusion, as it is a diffusion-based model.

AlignIF and its 'no-MStA' variant outperformed all competing methods on both test sets (Fig. 2a,b). On the benchmark test set, AlignIF achieved a recovery rate of 0.595 and a perplexity of 2.221, surpassing gRNAde (multistate; 0.444 and 2.692) and its no-MStA variant (0.574 and 2.314). On the independent test set, AlignIF reached a recovery rate of 0.487 and a perplexity of 2.723, again exceeding gRNAde (multistate; 0.390 and 3.072) and its no-MStA variant (0.481 and 2.764).

To investigate the impact of incorporating MStA, we conducted a direct comparison between AlignIF and AlignIF (no MStA). The results demonstrate that integrating MStA significantly enhances model performance, with more pronounced gains for RNAs that have deeper MStA depths, which correspond to a greater number of available MStA structures (Fig. 2c,d). This suggests that MStA enables AlignIF to model native RNA sequences more accurately. In head-to-head comparisons with gRNAde (multistate), AlignIF achieved better performance, particularly for longer RNAs (Fig. 2e,f).

Macro-F1 analysis confirmed AlignIF's high accuracy across all residue types, achieving optimal recovery rates while avoiding the overprediction of frequent nucleotides (Fig. 2g). In addition, our analysis highlighted a robust positive correlation between the average log probability of a sequence, conditioned on its structure, and its native recovery rate. Specifically, the Spearman rank correlation coefficient reached 0.72 (95% confidence interval (CI) 0.71–0.73) on the benchmark test set and 0.65 (95% CI 0.64–0.67) on the independent test set at a sampling temperature of 0.1, with 20 samples generated per structure (Fig. 2h). Compared with other methods, AlignIF consistently demonstrated stronger correlations across a range of temperatures (Extended Data Fig. 1). This correlation shows that AlignIF's internal probability scores strongly predict sequence recovery rates, making them a reliable statistical indicator of sequence nativeness.

To enhance the structural and sequential independence, we constructed new benchmark and independent test sets by removing any RNA whose structure has a TM score ≥0.40 or ≥0.35 relative to any training structures, and whose sequence similarity is ≥0.8 or ≥0.6 relative to any training sequences, respectively. We re-evaluated our trained models on each of these filtered test sets and report the recovery rate, perplexity and Macro-F1 score in Supplementary Tables 1 and 2. Notably, the MStA structures used during inference were also filtered to meet the same dissimilarity criterion. The performance confirms that AlignIF is stable across different dissimilarity thresholds of sequences and structures.

Capitalizing on the advances in RNA 3D structure prediction (for example, AlphaFold3[14], RhoFold+[15] and NuFold[19]), we tested 50 AlphaFold3-predicted RNAs from RNACentral[20], spanning 10–510 nucleotides, with an interval of 10 nucleotides. Maximum sequence similarity to the training set was 0.625 (Supplementary Fig. 1a), indicating limited overlap. AlignIF achieved 0.552 in recovery, 2.551 in perplexity and 0.549 in Macro-F1, corresponding to a 43.8% relative recovery gain over gRNAde (multistate), a 28% perplexity reduction versus gRNAde (single-state) and a 47.2% Macro-F1 gain over RiboDiffusion (Fig. 2i). Because these predicted structures have no MStA, AlignIF and AlignIF (no MStA) produced identical results. Notably, sequence recovery increased with AlphaFold3 prediction confidence, showing a positive correlation with the predicted TM score (pTM), with a Spearman rank correlation coefficient of 0.58 (Supplementary Fig. 1b). For a 470-residue partial 16S rRNA (URS00020ECA53), AlignIF reached 0.68 recovery and a self-consistency TM score of 0.93 (Fig. 2j and Supplementary Fig. 2), with pTM scores of 0.78 and 0.77 for native and designed sequences.

Evaluating the foldability of AlignIF-designed sequences

To evaluate whether the designed sequences fold into their target structures, we selected RNAs that do not interact with other biomacromolecules and contain no missing nucleotides, yielding 51 RNAs from the benchmark test set and 25 from the independent test set. Foldability was assessed using four self-consistency metrics: scTM, scRMSD, scGDT_TS and scMCC (Methods). The first three metrics compare the tertiary structures predicted from the designed sequences with the native target structures, whereas scMCC compares their secondary structures. For each designed sequence, its 3D structure was predicted using AlphaFold3, and its secondary structure was derived from the prediction. Because AlphaFold3 is less accurate for RNA 3D structure prediction than for proteins, we used native sequences to establish upper-bound reference metrics, providing a fair baseline given its inherent prediction error. Across both test sets, AlignIF approached these bounds and outperformed competing methods (Fig. 3a–d).

To mitigate the potential bias from relying on a single RNA 3D structure prediction method, we further evaluated the foldability using an independent predictor, RhoFold+. Because RhoFold+ is limited to single-chain RNAs, we excluded multichain structures from the original test sets, resulting in 32 RNAs from the benchmark test set and 24 from the independent set for this analysis (Supplementary Fig. 3). The results of RhoFold+ closely mirrored those obtained with AlphaFold3, reinforcing the conclusion that AlignIF achieves better foldability (Extended Data Fig. 2 and Supplementary Fig. 4).

We hypothesized that the good foldability of AlignIF arises not only from the high sequence recovery but also from learned folding patterns. To test this, we applied AlignIF to design a sequence for a riboswitch (8SA3)[21] in the test set, a functional RNA whose activity depends on its sophisticated 3D conformation. The designed sequence achieved a recovery rate of 0.64 and a scTM score of 0.76. For comparison, we generated two control sequences with identical recovery rates: Random1 (mutations only at unrecovered positions) and Random2 (random mutations matching AlignIF's unrecovered residue count). Both controls failed to fold properly (Random1, scTM 0.25; Random2, scTM 0.26; Fig. 3e), demonstrating that AlignIF's success depends not only on the sequence recovery but also on its learned ability to preserve critical structural determinants.

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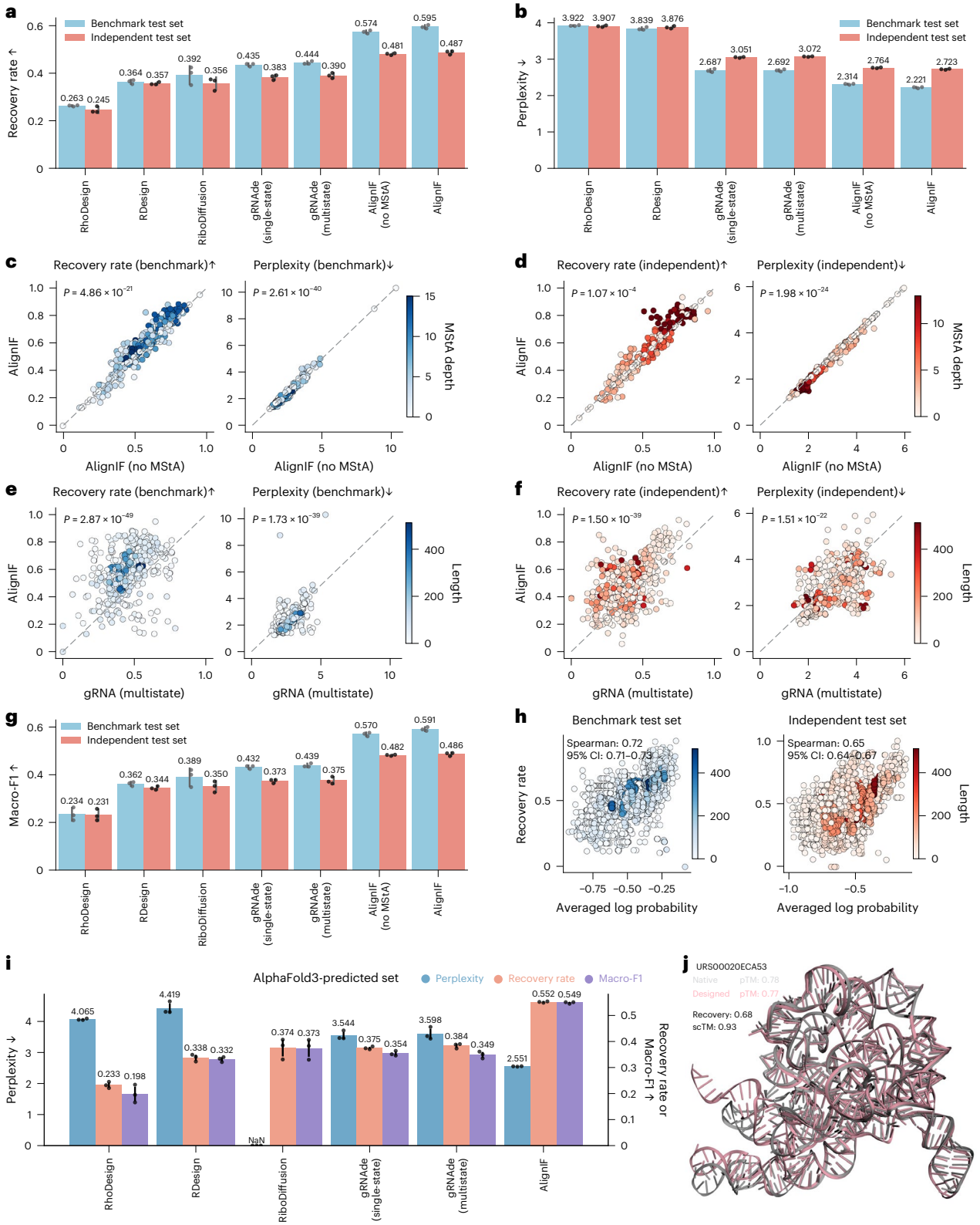

Fig. 2 | Performance evaluation of recovering native sequences. a, Sequence recovery rates (mean ± s.d. across three random seeds) across methods. **b**, Perplexity scores (mean ± s.d. across three random seeds), excluding RiboDiffusion, which is unable to calculate the perplexity. **c**,**d**, Head-to-head comparison of AlignIF versus AlignIF (no MStA) on the benchmark (**c**) and independent (**d**) test sets, evaluated across three random seeds. One-sided paired Wilcoxon signed-rank test *P* values are shown ($n = 474$). The color intensity denotes MStA depth. **e**,**f**, Head-to-head comparison of AlignIF versus gRNAde (multistate) on the benchmark (**e**) and independent (**f**) test sets, with one-sided paired Wilcoxon signed-rank test *P* values ($n = 558$). The color intensity denotes the sequence length. **g**, Macro-F1 scores (mean ± s.d. across three random seeds) of different methods. **h**, Correlation between the sequence recovery rate and average log probability (sampling temperature $T = 0.1$); Spearman coefficients and 95% CIs calculated via 5,000× bootstrap resampling are annotated. **i**, Performance metrics (perplexity, recovery rate, Macro-F1; mean ± s.d. across three random seeds) on AlphaFold3-predicted structures. **j**, Case study of a 470-residue partial 16S rRNA (URS00020ECA53), with predicted structures of native (gray) and AlignIF-designed (pink) sequences.

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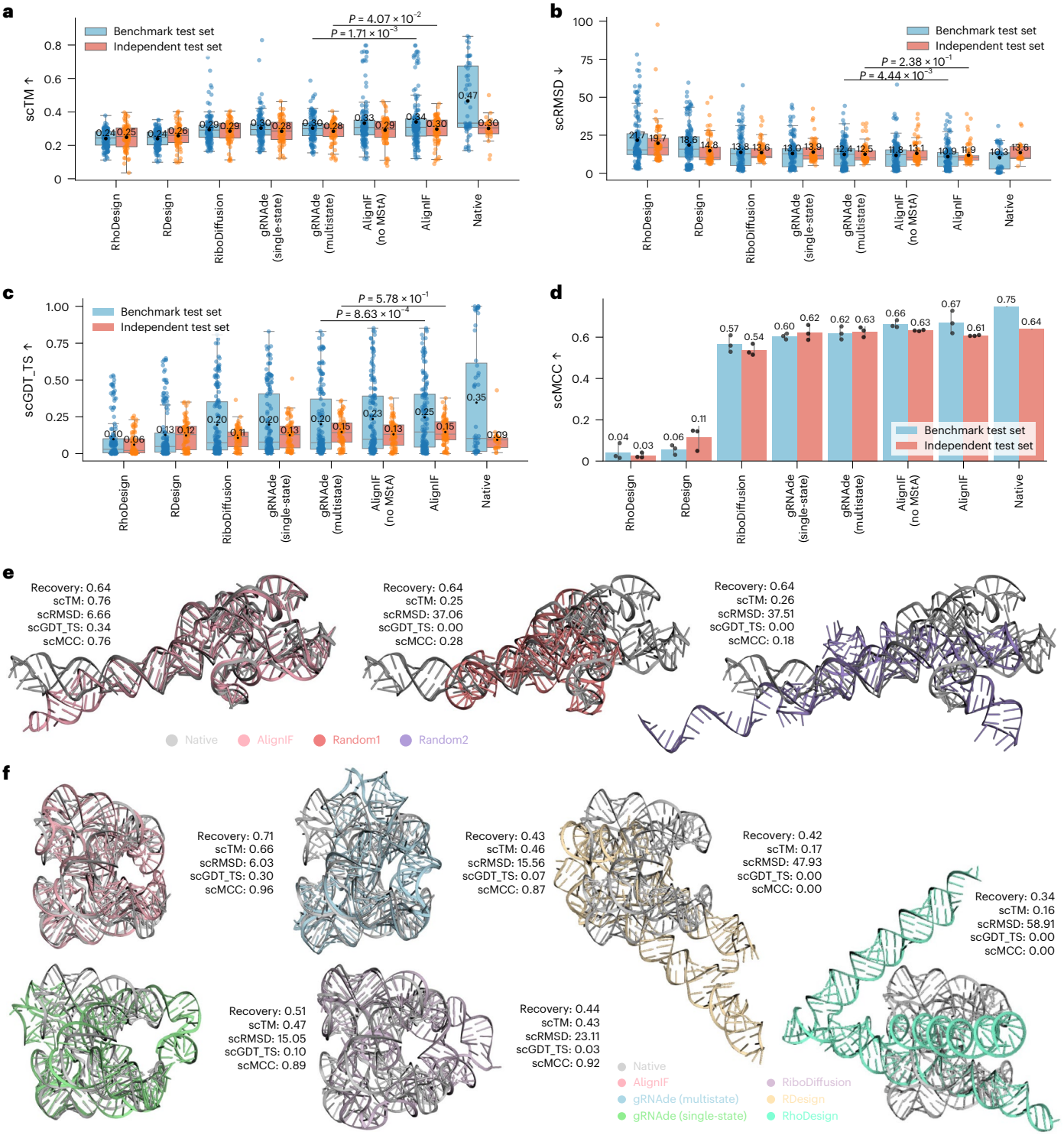

Fig. 3 | Foldability evaluation of RNA sequences designed by AlignIF and other methods using AlphaFold3. a–**c**, Tertiary-structure self-consistency metrics for RNA design methods across three random seeds: scTM score (**a**), scRMSD (**b**) and scGDT_TS (**c**). Box centers show medians, box limits show interquartile ranges (IQRs; 25th–75th percentiles), whiskers extend to the most extreme values within 1.5× IQR, and black points indicate means. Each box summarizes metric distributions across three independent random seeds, containing $n = 153$ values per method for the benchmark test set and $n = 75$ values per method for the independent test set. 'Native' columns show AlphaFold3-derived upper bounds from native sequences. One-sided paired Wilcoxon signed-rank test P values compare AlignIF with gRNAde (multistate). **d**, Secondary structure self-consistency measured by scMCC (mean ± s.d. across three random seeds). **e**, Riboswitch (8SA3) comparison. AlignIF (scTM 0.76) outperforms recovery-matched controls: Random1 (mutations only at unrecovered positions; scTM 0.25) and Random2 (random mutations matching the number of unrecovered nucleotides; scTM 0.26). **f**, Dimeric RNA nanocage (7JRS) case study comparing designed sequences across methods.

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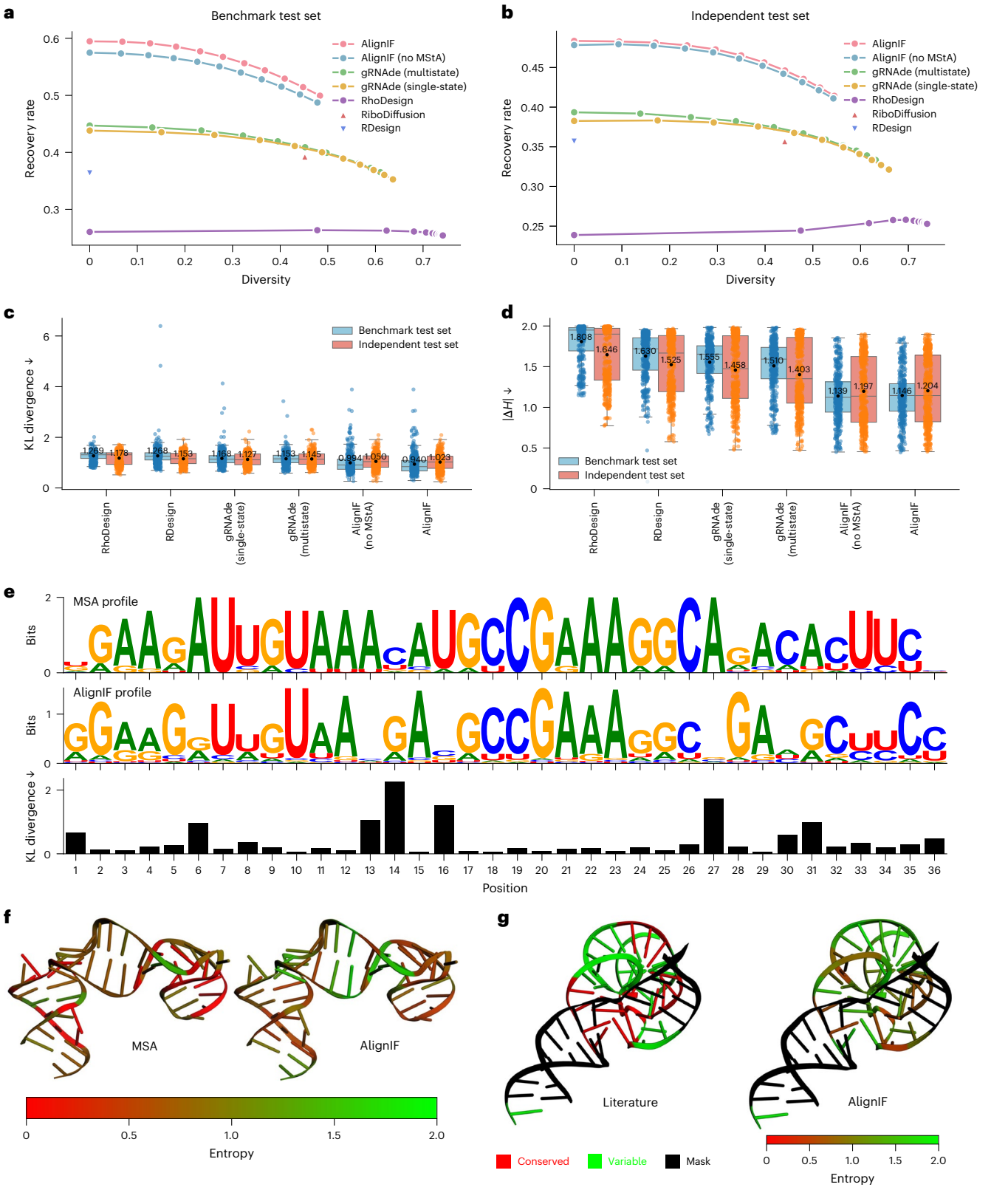

Fig. 4 | Analysis of evolutionary fidelity. a,**b**, Sequence diversity versus recovery rate, averaged across three random seeds, on the benchmark test set (**a**) and the independent test set (**b**). Circles represent sampling temperatures from 0 to 1.0 in 0.1 increments, increasing left to right. **c**,**d**, Model–profile comparison using KL divergence (**c**) and absolute entropy difference ($|\Delta H|$) (**d**) against MSA reference profiles. Box centers show medians, box limits show IQRs (25th–75th percentiles), whiskers extend to the most extreme values within 1.5× IQR, and black points indicate means. Each box summarizes three seeds, with $n = 411$ chains for the benchmark test set and $n = 492$ chains for the independent test set. **e**, Position-wise profile comparison for Clivia fluorogenic RNA: MSA (top), AlignIF (middle) and per-position KL divergence (bottom). **f**, Conservation analysis of RNase P RNA, showing agreement between AlignIF-predicted and MSA-derived conserved sites. **g**, Hammerhead ribozyme analysis comparing AlignIF predictions with literature-reported conserved nucleotides. Fixed substrate strand and paired nucleotides (black) were constrained during prediction.

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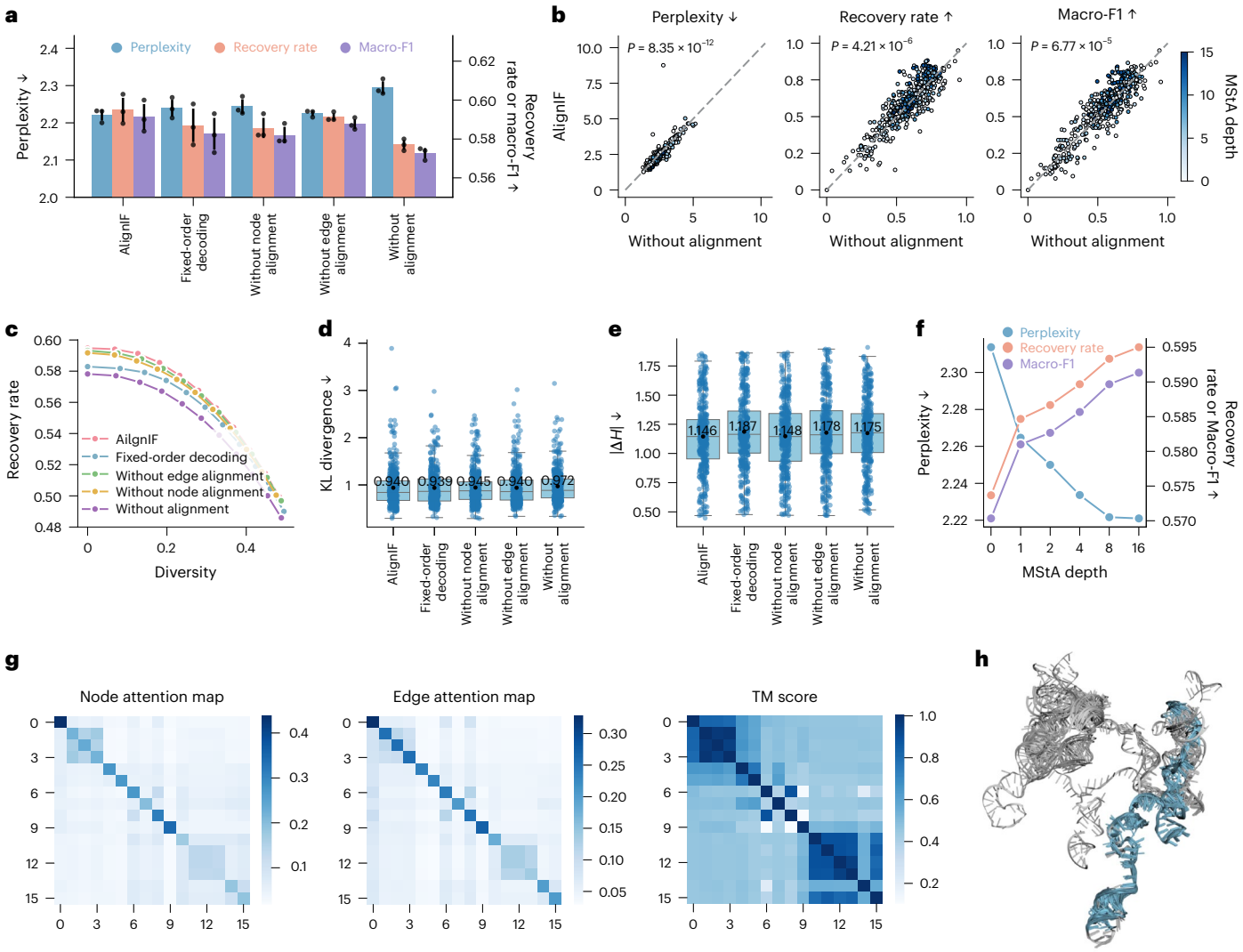

Fig. 5 | Ablation study of AlignIF. a, Ablation variants were compared on perplexity, recovery rate and Macro-F1 score (mean ± s.d. across three random seeds). **b**, AlignIF was compared head-to-head with the no-alignment variant across three seeds, with color intensity indicating MStA depth. One-sided paired Wilcoxon signed-rank test *P* values are shown ($n = 474$). **c**, The trade-off between sequence diversity and recovery rate was evaluated across ablation variants, averaged over three seeds. **d**,**e**, Output–profile similarity to MSA references was measured via KL divergence (**d**) and absolute entropy difference (**e**). Box centers show medians, box limits show IQRs (25th–75th percentiles), whiskers extend to the most extreme values within 1.5× IQR, and black points indicate means ($n = 411$ across three seeds). **f**, Impact of MStA depth on *Saccharomyces cerevisiae* U6 snRNA (chain 6 of 5MPS) in terms of perplexity, recovery rate and Macro-F1 score, averaged over three random seeds. **g**, Attention patterns for the same RNA: node attention map (left) and edge attention map (middle), averaged over nucleotides and network layers, alongside the TM-score map (right) for each structure aligned to the target. **h**, Aligned MStA structures, with blue nucleotides aligning to the target structure and gray nucleotides representing others.

As a case study, we further evaluated AlignIF on an artificial dimeric RNA nanocage (7JRS) in the test set, which includes complex structural elements such as kissing loops, branched kissing loops and T-junctions[22]. AlignIF successfully designed this intricate RNA nanoarchitecture, surpassing other methods across all self-consistency metrics (Fig. 3f). Notably, the sequence designed by AlignIF (no MStA) is identical to that designed by AlignIF, as no MStA exists for this artificial structure in the test set.

Capturing evolutionary rules and conservation patterns in RNA family sequences

We conducted a comparative analysis of the trade-off between sequence diversity and recovery rate across various methods. For autoregressive models, sequences were generated across a temperature range of 0–1.0, incremented in steps of 0.1. For non-autoregressive models, RDesign and RiboDiffusion, sequences were generated using their default configurations. As temperature increased, sequence diversity improved at the expense of the recovery rate. Among the compared models, AlignIF achieved the highest sequence diversity while maintaining better recovery performance (Fig. 4a,b).

While high sequence diversity and high native recovery are often competing objectives, AlignIF successfully achieves both. To understand this dual advantage, we analyzed its design principles in the context of biological and structural constraints. MSAs model evolutionary relationships within RNA families by capturing conservation and variation patterns among homologous sequences[23]. We compared model output profiles with MSA-derived sequence profiles using Kullback–Leibler (KL) divergence, restricting this analysis to chains with no missing nucleotides to ensure reliable MSA construction (Methods). The KL divergence thus measures agreement with broader evolutionary patterns rather than with a single native sequence. Across methods, AlignIF yielded profiles closest to the MSA-derived references (Fig. 4c and Extended Data Fig. 3a). For the Clivia fluorogenic RNA[24] (chain B of 8HZK) from the test set, AlignIF

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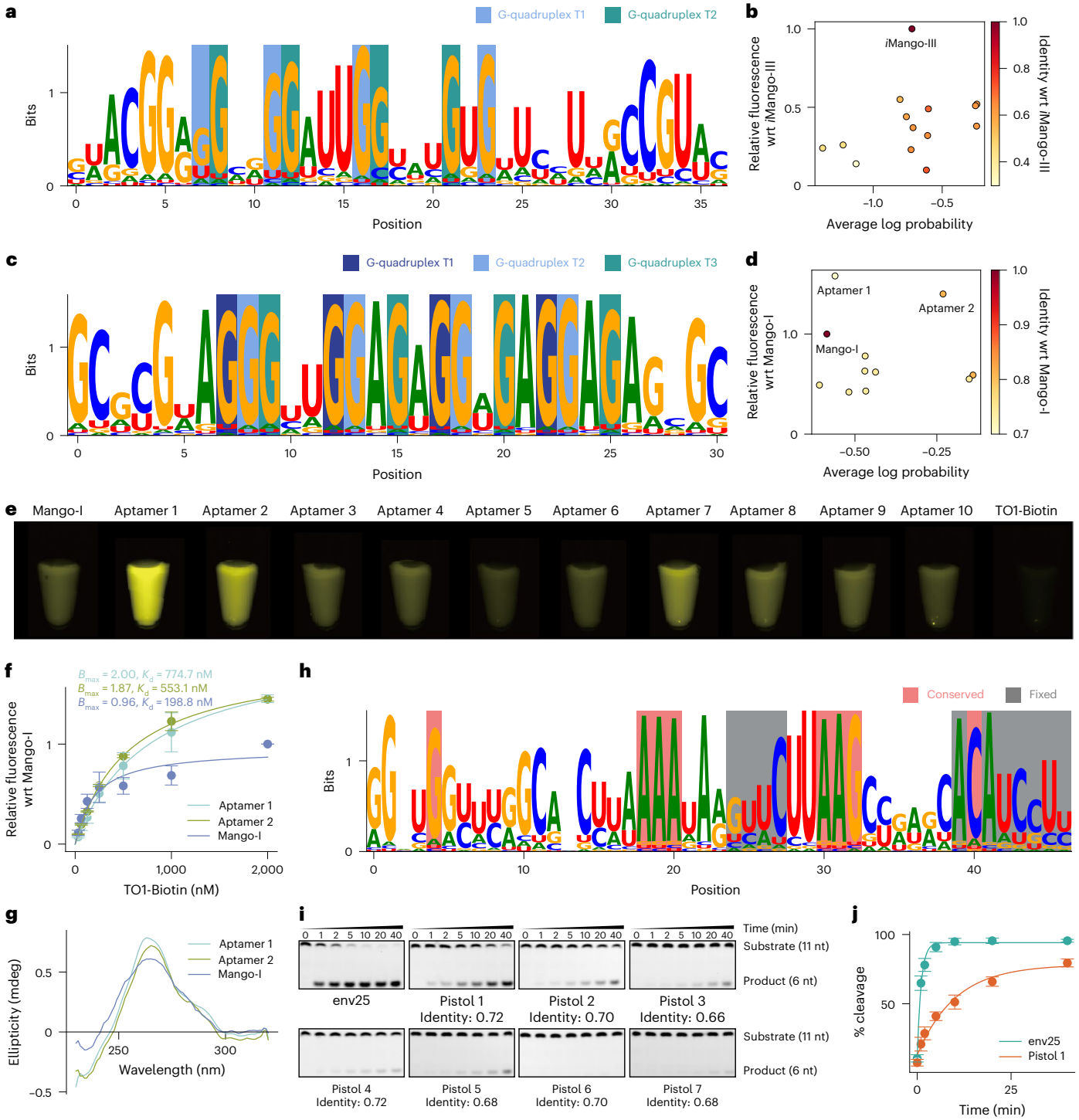

Fig. 6 | Experimental validation of AlignIF-generated fluorescent light-up RNA aptamers and pistol ribozymes. a, Sequence logo of AlignIF-generated aptamers based on the *i*Mango-III structure, with a two-tiered G-quadruplex motif highlighted by colored backgrounds. **b**, Relative fluorescence of generated aptamers with respect to *i*Mango-III. Color intensity indicates sequence identity to *i*Mango-III; the *x* axis shows the averaged log probability from AlignIF. **c**, Sequence logo of AlignIF-generated aptamers based on the Mango-I structure, with a three-tiered G-quadruplex motif highlighted by colored backgrounds. **d**, Relative fluorescence of generated aptamers with respect to Mango-I. Color intensity indicates sequence identity to Mango-I; the *x* axis shows the averaged log probability from AlignIF. Aptamers 1 and 2 exhibited ~1.6-fold and ~1.4-fold fluorescence enhancement over Mango-I, respectively. **e**, Fluorescence visualization of Mango-I and generated aptamers based on Mango-I. **f**, Measurement of K_d for aptamers 1 and 2, and Mango-I; 4 μM RNA was incubated with TO1-biotin at a series of final concentrations: 31.25, 62.5, 125, 250, 500, 1,000 and 2,000 nM. Fluorescence was normalized to Mango-I with 2,000 nM TO1-biotin. Normalized fluorescence values were fitted using Prism 9, GraphPad. Error bars show the standard error of the mean of three independent replicates. **g**, CD spectroscopy of aptamers 1 and 2, and Mango-I; 2 μM RNA incubated with 4 μM TO1-biotin, 140 mM KCl and 1 mM $MgCl_2$ in 10 mM NaH_2PO_4 (pH 7.2) buffer. CD spectra were recorded at 25 °C from 230 to 320 nm using a 0.5-mm path-length quartz cuvette, with three independent replicates. **h**, Sequence logo of generated ribozymes based on env25. Fixed nucleotides are indicated by gray shading, and highly conserved positions are shown in pink. **i**, Representative urea-PAGE gel showing generated ribozymes performing single-turnover of substrate RNA cleavage. The experiment was repeated independently three times with similar results. **j**, Pre-steady-state bimolecular reaction of 50 nM substrate with 1 μM pistol 1 and env25 (23 °C, pH 7.5, 10 mM Mg^{2+}; mean ± s.d. across three independent experimental replicates), to measure k_{obs}. wrt, with respect to.

recovered native sequences while matching the family's evolutionary rules (Fig. 4e).

To evaluate how accurately the models capture conservation patterns in RNA families, we computed the entropy of profiles, where low-entropy sites represent conserved positions. By quantifying the absolute entropy difference ($|\Delta H|$) between model outputs and MSA-derived profiles, we found AlignIF and AlignIF (no MStA) achieved the smallest deviations, performing comparably to each other, demonstrating their better ability to identify both conserved and flexible regions (Fig. 4d and Extended Data Fig. 3b). For example, in the case of RNase P RNA[25] (chain B of 5Y7M) from the test set, AlignIF's predicted conserved sites showed strong agreement with those derived from MSAs (Fig. 4f and Supplementary Fig. 5). Furthermore, in a hammerhead ribozyme (8YDC) from the test set analysis, where we constrained the substrate strand and its paired nucleotides, AlignIF's predictions aligned precisely with literature-reported conserved positions[26]. The results demonstrate AlignIF's ability to identify conserved and flexible regions accurately based on 3D RNA structure (Fig. 4g and Supplementary Fig. 6). In summary, these results demonstrate AlignIF's capacity to integrate structural constraints with evolutionary principles, enabling diverse yet biologically plausible RNA designs.

Ablation study and module contributions in AlignIF

To assess the contributions of each module in AlignIF, we performed an ablation study on the benchmark dataset using several model variants. The MStA backbone encoder concurrently updates nodes and edges across graphs, and we investigate the impact of cross-graph node and edge updates. In addition, our model incorporates a random-ordered autoregressive sequence decoder, a design choice that helps mitigate overfitting during training. The ablation study includes the following variants: (1) fixed-order decoding: replaces the random decoding order with a fixed-order decoder during training; (2) without edge alignment: removes cross-graph edge updates; (3) without node alignment: removes cross-graph node updates; and (4) without alignment: eliminates both node and edge alignment.

Evaluation based on perplexity, recovery rate and Macro-F1 score confirms that all modules are essential (Fig. 5a). Among the ablated variants, the model without alignment performs the worst across all metrics. A head-to-head comparison between AlignIF and the variant without alignment reveals that cross-graph node and edge updates significantly improve the design capability, particularly for samples with larger MStA depth (Fig. 5b). We also evaluated whether random decoding order mitigates the overfitting and exposure bias (Supplementary Fig. 7).

Our analysis of the trade-off between the sequence diversity and recovery rate shows that ablating either node alignment or edge alignment in isolation results in only minor performance degradation. By contrast, removing both components simultaneously results in a sharp decline in the performance (Fig. 5c). Compared with MSA profiles, the model without the alignment yields the highest KL divergence, while the fixed-order decoding variant shows the largest absolute entropy difference. These demonstrate the critical roles of both alignment and random decoding order in capturing RNA family evolutionary patterns.

Motivated by the inherent flexibility of the sugar ring, AlignIF introduces two additional sugar angles as node features (Supplementary Fig. 8b). To assess their impact, we compared AlignIF with a variant that excludes the two additional sugar angles in feature extraction. The results show that including these angles helps AlignIF better model conformational changes within the sugar ring, improving the design performance (Supplementary Fig. 9). This underscores the importance of the two features in enhancing RNA sequence generation.

To examine the effect of MStA depth, we used *Saccharomyces cerevisiae* U6 snRNA (chain 6 of 5MPS), which had the deepest MStA in the test set. The MStA set is constructed by greedily selecting structures ranked by their TM scores against the target structure, thereby prioritizing the inclusion of the most structurally similar models. AlignIF consistently improves as the MStA depth increases (Fig. 5f), demonstrating the added value of similar structures. The maximum MStA depth is set to 16 owing to the limited size of the available MStA set (Supplementary Fig. 10). Even without MStA, AlignIF achieved a recovery rate of 0.574, a perplexity of 2.314 and a Macro-F1 score of 0.570 on the benchmark set, outperforming gRNAde (multistate), which achieved 0.444, 2.692 and 0.439, respectively. This confirms the strength of the architecture and structural features even in the absence of MStA homologs.

To further investigate how AlignIF leverages MStA information, we analyzed its attention patterns across incorporated structures. Averaged self-attention maps, computed over nucleotides and network layers for both nodes and edges, show clear correspondence with the TM-score map (Fig. 5g). The aligned MStA structures are also presented, with blue nucleotides indicating nucleotides aligned to the target structure and gray nucleotides representing others (Fig. 5h).

Finally, we evaluated the computational overhead of MStA and tested whether Gaussian backbone perturbations could substitute for aligned homologous structures. AlignIF's memory cost increased with MStA depth because cross-graph attention scales with the number of aligned structures, whereas graphics processing unit (GPU) parallelization kept runtime relatively stable (Supplementary Fig. 11). Replacing MStA with Gaussian backbone perturbations produced AlignIF (noise), which performed similarly to the no-alignment model and failed to reproduce the MStA gains (Supplementary Fig. 12), demonstrating that the benefit of MStA arises from evolutionarily conserved and physically plausible structural variation rather than arbitrary noise.

Wet-lab experimental validation for fluorescent light-up RNA aptamer design

Following in silico benchmarking, we further use wet-lab experiments to validate AlignIF by generating RNA sequences based on known functional RNA structures. This allowed us to assess whether AlignIF could produce sequences that not only fold into structures similar to the target but also retain biological functions.

We first focused on fluorescent light-up RNA aptamers, which bind small molecules and emit fluorescence, making them valuable tools for RNA imaging and detection in live cells[27]. Mango aptamers, which bind TO1-biotin (a thiazole orange derivative), have well-characterized crystal structures and binding mechanisms[28–31]. We selected *i*Mango-III (chain B of 6E8U), which exhibits 13% higher brightness than Mango-III[30], as our first design target. To prevent the bias, we retrained AlignIF on RNA structures with low structural similarity (<0.45) to *i*Mango-III. Analysis of the retrained model's structural output profile revealed that AlignIF preferentially generated sequences containing the two-tiered G-quadruplex, a critical structural motif for fluorophore binding (Fig. 6a). We randomly sampled and synthesized 13 aptamers with sequence identities to *i*Mango-III ranging from 0.30 to 0.73. All generated aptamers showed fluorescence activity in the presence of TO1-biotin. Among them, three aptamers exhibited fluorescence intensities exceeding half that of *i*Mango-III (Fig. 6b). Although none of them outperformed *i*Mango-III in brightness, all aptamers retained at least one-tenth of their fluorescence intensity, demonstrating AlignIF's capability to generate functional fluorescent light-up RNA aptamers.

Although none exceeded *i*Mango-III in brightness, this is expected because *i*Mango-III was previously optimized by competitive in vitro selection from 4.1×10^6 Mango-III variants[30]. We then targeted wild-type Mango-I (chain A of 5V3F)[28] after retraining on structures dissimilar to 5V3F (chain A or B). The model output profile retained the three-tiered G-quadruplex functional core (Fig. 6c). Ten designs with 0.70–0.80 identity to Mango-I were all active in the presence of TO1-biotin and exhibited fluorescence intensities at least 0.4-fold that of Mango-I. Two designs, aptamer 1 and aptamer 2, were brighter than Mango-I by 1.6-fold and 1.4-fold (Fig. 6d,e). Designs

generated without MStA remained functional but were significantly dimmer (Supplementary Fig. 13), confirming that MStA enhances functional performance.

To investigate the mechanism behind this fluorescence enhancement, we measured the dissociation constants (K_d) of these aptamers. Surprisingly, aptamers 1 and 2 had higher K_d values than Mango-I (3.9-fold and 2.8-fold, respectively), indicating reduced binding affinity for TO1-biotin (Fig. 6f and Extended Data Fig. 4a,b). Nevertheless, the observed fluorescence enhancement can be attributed to their higher B_{max} values, which suggest an increased fluorophore capacity. Specifically, aptamer 1 and aptamer 2 accommodated ~2.1-fold and ~2.0-fold more TO1-biotin, respectively, compared with Mango-I. Notably, two additional aptamers were identified that displayed improved binding affinity, with K_d values 3.8-fold and 2.5-fold lower than that of Mango-I, respectively (Extended Data Fig. 4b).

To further investigate the mechanism underlying fluorescence enhancement, we performed circular dichroism (CD) spectroscopy. Because RNA G-quadruplexes exhibit topology-dependent CD signatures, CD spectra can provide insight into their folding conformations. In general, parallel G-quadruplexes show a positive band at ~260 nm and a negative band at ~240 nm. By contrast, antiparallel G-quadruplexes are characterized by a prominent positive band at ~295 nm and a negative band near 260 nm (ref. 32). Although the co-crystal structure of Mango-I bound to TO1-biotin revealed a three-tiered G-quadruplex with mixed parallel and antiparallel connectivity[28], the CD spectra of aptamer 1, aptamer 2, and Mango-I exhibited a strong positive band at 265 nm and a negative band at 235 nm (Fig. 6g), a pattern more characteristic of a parallel G-quadruplex topology. This apparent discrepancy may reflect differences in buffer conditions between crystallography and CD spectroscopy, which could favor distinct G-quadruplex conformations. Aptamers 1 and 2 showed slightly stronger 265-nm signals than Mango-I with TO1-biotin alone (Extended Data Fig. 4c), and these signals became substantially stronger after Mg^{2+} and K^+ addition (Fig. 6g). Because the strong positive CD signal near 260 nm is associated with a greater likelihood of G-quadruplex formation and increased G-tetrad stacking stability[33,34], aptamers 1 and 2 probably fold into stable G-quadruplexes more efficiently than Mango-I, contributing to their fluorescence enhancement.

To assess whether computational metrics could prioritize experimentally active designs, we retrospectively analyzed average log probability, the scTM score of AlphaFold3-predicted structures, scMCC of secondary structures derived from AlphaFold3 predictions, and scMCC of secondary structures predicted by RibonanzaNet[35] across the experimentally tested designs. These metrics showed target-dependent associations with activity (Supplementary Fig. 14), indicating that no universal target-agnostic ranking strategy could be identified from the current experiments.

In summary, our findings demonstrate that AlignIF can effectively design functional fluorescent light-up RNA aptamers. The observed fluorescence enhancement in the two designed aptamers can be attributed to increased fluorophore binding capacity and the formation of more stable G-quadruplex structures, highlighting AlignIF's capability in exploring functional RNA landscapes.

Experimental validation of motif-guided pistol ribozyme design by AlignIF

Small self-cleavage ribozymes are an important class of functional RNAs. They catalyze site-specific internal transesterification, involving the ribose 2′-hydroxyl group adjacent to the scissile phosphate[36]. Due to their precise cleavage activity, these ribozymes serve as valuable tools for gene-silencing therapies[37–39]. The field of small self-cleavage ribozymes has been invigorated by the recent discovery of the twister, twister-sister, pistol and hatchet ribozymes[36]. However, the complex structures of ribozymes and the scarcity of available PDB structures of ribozymes pose challenges for rational design.

We focused on the pistol ribozyme env25 (chain A of 5K7C), which adopts a compact tertiary structure stabilized by an embedded pseudoknot[40]. Ribozymes rely on their binding arms to recognize specific RNA substrates and catalyze the cleavage of phosphodiester bonds[41,42]. To ensure the specificity toward desired substrates, we fixed the binding arm of the pistol ribozyme during sequence generation using AlignIF. Notably, the crystal structure of 5K7C contains a deoxyguanosine (dG) substitution at the G–U cleavage site to prevent rapid and unintended self-cleavage, introducing a structural discrepancy near the active site. To account for this, we also fixed nucleotides G40 and G42, which are proximal to the cleavage site, when designing new ribozyme sequences with AlignIF.

After retraining AlignIF, the resulting sequence profile recovers highly conserved nucleotides[43] (Fig. 6h), despite the absence of any pistol ribozyme in the training set. We experimentally assessed the RNA cleavage activity of the seven generated sequences by measuring their pseudo-first-order rate constants (k_{obs}) (Fig. 6i). All ribozymes exhibited weak cleavage activity, with pistol 1 exhibiting 11% of the k_{obs} value of the wild-type env25 (Fig. 6j and Extended Data Fig. 4d). The reduced activity may arise from structural deviations between the crystal structure and the real biologically active conformation, as well as limited structural information available regarding the dynamic cleavage process. Nevertheless, our results demonstrate that AlignIF can successfully generate functional self-cleaving ribozymes, albeit with room for further optimization.

Discussion

AligniF achieves promising performance in functional RNA design, but several practical limitations still remain. First, candidate selection for experimental validation is unresolved. The retrospective analyses indicate that activity-associated metrics are target dependent, underscoring that no universal target-agnostic ranking strategy has yet been integrated into AlignIF. This challenge is consistent with prior RNA design pipelines, in which downstream filters are often tailored to specific RNA classes or design settings. For example, RhoDesign is tailored to SELEX-derived aptamers and relies on motif information from prior experiments[8], whereas NA-MPNN and gRNAde use SHAPE-reactivity-informed consistency and OpenKnot-related criteria that are especially relevant to pseudoknot-focused design[44,45]. Developing robust target-agnostic ranking strategies therefore remains an important direction for future work. Second, AlignIF's MStA modeling creates increasing memory overhead, although runtime is mitigated by GPU parallelization. Pruning low-TM-score structures or distilling the MStA encoder could reduce computational overhead.

The pistol ribozyme results also highlight current limits. None of the designs improved the cleavage activity, probably because the crystal structure differs from the biologically active conformation and lacks dynamic cleavage-state information. Additional structural data, especially intermediate or functional states, and integration of sequence-level evolutionary constraints could improve ribozyme optimization[46].

Future development of AlignIF should therefore focus on making the framework more function-aware, data-integrative and computationally efficient. Incorporating experimental constraints from chemical probing, such as SHAPE, or structural information from cryo-electron microscopy density maps could improve the reliability of input ensembles, especially for novel RNA folds with limited homologous information. Active learning strategies may further improve the design performance for rare RNA families by iteratively selecting high-value candidates for experimental validation and using the resulting data to refine the model. In the longer term, extending AlignIF beyond RNA-only systems to RNA–protein complexes could broaden its applications to engineered ribonucleoprotein assemblies, CRISPR-associated systems and other synthetic biology platforms. Together, these directions would move AlignIF from

structure-conditioned sequence generation toward more reliable function-aware RNA engineering.

Methods

Datasets

Benchmark dataset. We constructed the benchmark dataset from 3D RNA structures in the BGSU RNA 3D Hub using a temporal cutoff of 1 May 2024. Structures were required to have resolution better than 4.0 Å and sequence lengths of 16–512 nucleotides, covering most functional RNAs, including small noncoding RNAs, signal recognition particle RNAs, aptamers, riboswitches and ribozymes[47–51]. This yielded 7,932 structures. The sequence length distribution is shown in Supplementary Fig. 15. To prevent information leakage, structures were clustered by TM score, with RNAs exceeding 0.45 assigned to the same cluster. The resulting 499 clusters were split into training, validation and testing sets comprising 399 clusters (7,669 RNAs), 50 clusters (129 RNAs) and 50 clusters (158 RNAs), respectively. Modified nucleotides were replaced with canonical counterparts when available.

Independent test set. To establish a strictly independent test set for model evaluation, we applied a dual-filter strategy:

(1) Temporal isolation: RNA structures were collected exclusively from entries released between 1 May 2024 and 1 January 2025.
(2) Structural dissimilarity control: We applied a TM-score threshold of 0.45 to exclude any RNAs that exhibited structural similarity to those in the benchmark dataset, ensuring no overlap between the two datasets.

After applying these filters, the final independent test set consisted of 186 RNA structures.

AlphaFold3-predicted dataset for in silico predicted structures. To evaluate AlignIF on computationally predicted RNA structures, we built a synthetic dataset using AlphaFold3. We randomly selected 50 RNACentral sequences with lengths from 20 to 510 nucleotides in increments of 10, enabling the assessment of modern predicted RNA structures.

MStA construction

To identify MStA members, we used US-align to obtain global TM-scores and local alignment pairs. Structures with TM score >0.45 relative to the target were included in MStA, reflecting conserved global folds and potential functional homology. Unless stated otherwise, MStA depth, the number of structures used during inference, was set to all available structures in MStA.

This MStA threshold matches the clustering thresholds (TM score 0.45) used for the benchmark and independent datasets, ensuring consistent evaluation and keeping RNAs in the test sets and their MStAs structurally distinct from training and validation structures.

MSA construction

To evaluate evolutionary fidelity, we constructed MSAs for targets without missing nucleotides, following RNAcmap[52]. Homologs were first retrieved with BLASTN[53] against the RNACentral and NCBI nucleotide databases (E-value threshold 0.001, line length 1,000, maximum number of alignments 50,000). A consensus secondary structure was predicted with RNAfold[54] and used with the homologs to build a covariance model (CM) using INFERNA[55] cmbuild. The CM was calibrated with cmcalibrate and used for a sensitive cmsearch against both databases (E-value threshold 10.0). Retrieved homologs were aligned to generate the final MSA for downstream evolutionary analyses.

3D RNA structure representation

Graph construction. AlignIF represents RNA structures as graphs whose nodes are nucleotides and whose edges encode spatial relationships. Each nucleotide position is defined by its C4′ atom, and edges connect nodes within 20 Å. We evaluated cutoffs of 15 Å, 20 Å and 25 Å (Extended Data Fig. 5). While 25 Å slightly improved performance, it increased average node degree from 21.44 to 32.14 and substantially increased memory use and computation. Thus, 20 Å balances long-range interaction capture and tractability. For MStA structures, AlignIF constructs one graph per structure and uses cross-graph modeling to capture MStA-specific features.

Node features. The RNA representation must remain invariant to global rotations and translations, as such transformations do not alter the underlying sequence. AlignIF constructs a local coordinate system for each nucleotide based on the O4′, C4′ and C3′ atoms in the sugar ring. The x axis is defined as the bisector of the O4′–C4′ and C3′–C4′ bond directions. The y axis is determined by the cross product of the O4′–C4′ and C3′–O4′ vectors, and the z axis is obtained as the cross product of the x and y axes (Supplementary Fig. 8a).

Within this frame, we compute directional vectors from C4′ to the phosphate (P) atom and the base-specific nitrogen atom (N1 for pyrimidines and N9 for purines). We also include nine dihedral angles: six backbone torsions, one base-orientation dihedral and two sugar-ring angles (C2′–C1′–O4′–C4′ and O4′–C1′–C2′–O2′), capturing sugar conformations (Supplementary Fig. 8b). Dihedral angles are represented by sine and cosine values. Spatial orientation between consecutive nucleotides is encoded by two rotation matrices aligning the current local frame with preceding and succeeding frames, compactly represented as unit quaternions.

Edge features. Spatial relationships between neighboring nucleotides are described using distances from the P, C4′ and N atoms of the neighbor to the current C4′ atom, as well as the neighbor N–N distance (Supplementary Fig. 8c). Distances are encoded with radial basis functions, and relative rotations between local coordinate systems are represented as unit quaternions.

AlignIF architecture

MStA backbone encoder. After representing MStA structures as multiple geometric graphs, nodes across different graphs are aligned using US-align. The edges between corresponding nodes are also aligned accordingly. Within each encoder block, AlignIF first performs independent message passing on each geometric graph to update node features:

$$\mathbf{m}_{\mathrm{i}}^{\mathrm{k}} = \frac{1}{|\mathcal{N}(\mathrm{i})|} \sum_{\mathrm{j}\in\mathcal{N}(\mathrm{i})} \mathrm{NodeMLP}(\mathbf{v}_{\mathrm{i}}^{\mathrm{k}}, \mathbf{v}_{\mathrm{j}}^{\mathrm{k}}, \mathbf{e}_{\mathrm{ij}}^{\mathrm{k}}) \tag{1}$$

where $\mathcal{N}(i)$ denotes the set of neighbors of the nucleotide i, the superscript k represents the kth structure in MStA, and $\mathbf{v}$ and $\mathbf{e}$ denote node and edge features, respectively.

Next, the updated node feature $\mathbf{m}_{\mathrm{i}}^{\mathrm{k}}$ is processed using a self-attention mechanism across all K structures in MStA. This allows integration of information across all aligned graphs while preserving permutation invariance, owing to the position-agnostic nature of the attention mechanism within each graph:

$$w_i^{kn} = \frac{\exp\left(\mathrm{Att}\left(\mathbf{m}_i^k, \mathbf{m}_i^n\right)\right)}{\sum_{o=1}^{K} \exp\left(\mathrm{Att}\left(\mathbf{m}_i^k, \mathbf{m}_i^o\right)\right)} \tag{2}$$

$$\mathrm{Att}\left(\mathbf{m}_{\mathrm{i}}^{\mathrm{k}}, \mathbf{m}_{\mathrm{i}}^{\mathrm{n}}\right) = \mathbf{m}_{\mathrm{i}}^{\mathrm{k}} W_{\mathrm{Q}} \left(\mathbf{m}_{\mathrm{i}}^{\mathrm{n}} W_{\mathrm{K}}\right)^{\top} \tag{3}$$

$$\hat{\mathbf{v}}_i^k = \sum_{n=1}^{K} w_i^{kn} \mathbf{m}_i^n W_{\mathrm{node}}, \tag{4}$$

where $w_{\mathrm{i}}^{\mathrm{kn}}$ is the attention weight between the kth and nth structures for node i, $\hat{\mathbf{v}}_{\mathrm{i}}^{\mathrm{k}}$ is the updated node feature for nucleotide i in the kth structure, and W_{Q}, W_K and W_{node} are learnable projection matrices.

In addition to updating edge features, we similarly update edge features:

$$\tilde{\mathbf{m}}_{ij}^{k} = \mathrm{EdgeMLP}\left(\hat{\mathbf{v}}_{i}^{k}, \hat{\mathbf{v}}_{i}^{k}, \mathbf{e}_{ij}^{k}\right) \tag{5}$$

$$\tilde{w}_{ij}^{kn} = \frac{\exp\left(\mathrm{Att}\left(\tilde{\mathbf{m}}_{ij}^{k}, \tilde{\mathbf{m}}_{ij}^{n}\right)\right)}{\sum_{o=1}^{K}\exp\left(\mathrm{Att}\left(\tilde{\mathbf{m}}_{ij}^{k}, \tilde{\mathbf{m}}_{ij}^{o}\right)\right)} \tag{6}$$

$$\hat{\mathbf{e}}_{ij}^{k} = \sum_{n=1}^{K} \tilde{w}_{ij}^{kn}\tilde{\mathbf{m}}_{ij}^{n}W_{\mathrm{edge}}, \tag{7}$$

where $\hat{\mathbf{e}}_{ij}^{k}$ denotes the updated edge features between nodes i and j in the kth structure, and W_{edge} is a learnable projection matrix for edge features.

The MStA backbone encoder iteratively refines node and edge representations through alternating intragraph message passing and intergraph attention. Because node and edge representations are SE(3)-invariant, an equivariant architecture would add complexity and computational cost without benefit. The encoder therefore uses an SE(3)-invariant GNN architecture, enabling cross-structure modeling across MStA.

Sequence decoder. The final encoder representations encode MStA information while remaining invariant to structural ordering within the MStA. To avoid ambiguity from multiple structural inputs, the decoder uses only the target graph, whose representation is informed by other MStA structures. The decoder autoregressively predicts probabilities over A, G, C and U for each node. Inspired by ProteinMPNN, it uses order-agnostic decoding by randomly sampling base-sequence permutations (Supplementary Algorithm 1), improving design flexibility and reducing overfitting to fixed sequence orders.

Training details

The MStA backbone encoder and sequence decoder each comprise three layers. Node and edge hidden features are represented with a dimensionality of 128. The model is trained using a mini-batch size of 64 RNAs and a learning rate of 1×10^{-4}, on a single NVIDIA GeForce RTX 4090 D GPU, with the Adam optimizer[56]. During training, label smoothing with a rate of 0.1, dropout with a rate of 0.2 and an exponential moving average with a decay rate of 0.999 are applied to improve the generalization and training stability.

AlignIF is trained in a self-supervised fashion to reconstruct native RNA sequences. The objective is to minimize the cross-entropy loss between the predicted nucleotide probabilities and ground-truth nucleotides at each node. Teacher forcing is employed during training, feeding the decoder the true previous token rather than its own prediction to encourage faster, more stable convergence.

Comparison with other models

To demonstrate the advantage of AlignIF, we compare it with state-of-the-art models for 3D structure-based RNA sequence design approaches RDesign[5], RiboDiffusion[6], RhoDesign[8] and gRNAde (single-state and multistate versions)[7].

- RDesign employs a hierarchical, data-efficient framework that incorporates contrastive learning at both the cluster and sample levels. It embeds RNA structures into a hyperspherical space, enhancing design performance especially under data-limited conditions.
- RiboDiffusion is a generative diffusion model for structure-based RNA sequence design that learns the conditional distribution of RNA sequences given 3D backbone structures. It treats sequences as one-hot vectors and operates by adding and removing Gaussian noise in the continuous space.
- RhoDesign combines a GVP encoder, a Transformer-based encoder–decoder architecture, and both secondary and tertiary structure information. It extracts structural features using the N1 and C2 atoms across all nucleotides, regardless of whether they are pyrimidines or purines. However, due to systematic geometric differences between purines and pyrimidines, the spatial positions of these atoms implicitly encode nucleotide identity. This may allow the model to rely on trivial structural cues to distinguish base types. To ensure fairness, we modified the feature extraction process by using topologically corresponding atom pairs: N1 and C2 for pyrimidines, and N9 and C4 for purines.
- gRNAde leverages either a single-state or multistate GNN in conjunction with autoregressive decoding to generate candidate RNA sequences conditioned on one or more 3D backbone structures, which employs pooling to integrate multiple structural information, supporting both static and conformationally dynamic design tasks.

We retrained these models without modifying any hyperparameters or architecture, except for adjusting the number of training epochs (dynamically determined through early stopping based on validation loss) and correcting the implicit nucleotide identity encoding in RhoDesign. The training and validation loss curves for these methods are shown in Supplementary Fig. 16. Supplementary Table 3 provides a comprehensive overview of computational costs with the model sizes (total parameters) and training wall-clock times for all methods.

Nucleotide profile estimation

Except for RiboDiffusion, all baseline models produce a probability distribution over nucleotides at each position in the RNA sequence. Among these, AlignIF, gRNAde and RhoDesign are autoregressive models. Their joint distribution over an RNA sequence of length L factorizes as

$$\mathrm{p}(x_1, x_2, \ldots, x_L) = p(x_1)\prod_{\mathrm{i}=2}^{\mathrm{L}} p(x_i|x_{<i}) \tag{8}$$

where x_{i} denotes the nucleotide at position i, and $x_{<i} = (x_1, \ldots, x_i)$ represents the preceding context.

For such models, the marginal probability (the nucleotide profile) at position i is obtained by marginalizing over all possible preceding sequences:

$$p(x_i) = \int p(x_{1:i})\mathrm{d}x_{<i} = \int p(x_{<i})p(x_i|x_{<i})\mathrm{d}x_{<i} = \mathrm{E}_{x_{<i}\sim p(x_{<i})}[p(x_i|x_{<i})]. \tag{9}$$

In practice, this expectation is approximated by sampling multiple trajectories for the prefix $x_{<\mathrm{i}}$, and averaging the resulting conditional probabilities $p(x_i|x_{<i})$.

Similarly, the joint probability of base pairs at positions i and j is computed as

$$\begin{aligned} p(x_i, x_j) &= \mathrm{p}(x_j|x_i)\,p(x_i) \\ &= \mathrm{E}_{x_{<i}\sim p(x_{<i})}\left[p(x_i|x_{<i})\int p(x_{i+1:j}, |, x_{<i})\,dx_{i+1:j-1}\right] \\ &= \mathrm{E}_{x_{<i}\sim p(x_{<i})}\left[p(x_i, |, x_{<i})\int p(x_{i+1:j-1}|x_{<i})\,p(x_j|x_{<j})\,dx_{i+1:j-1}\right] \\ &= \mathrm{E}_{x_{<i}\sim p(x_{<i})}\left[p(x_i|x_{<i})\,\mathrm{E}_{x_{i+1:j-1}\sim p(x_{i+1:j-1}|x_{<i})}\left[p(x_j|x_{<j})\right]\right] \end{aligned} \tag{10}$$

In this study, we use 20 sampled trajectories to approximate these expectations.

By contrast, the non-autoregressive model RDesign directly outputs marginal probability distributions for each position. These can be used to construct the nucleotide profile without the need for marginalization over preceding contexts.

Performance metrics

To benchmark AlignIF against baseline models, we used the following evaluation metrics.

- Recovery rate. This metric measures the fraction of nucleotides in the designed RNA sequence that correctly match the corresponding positions in the natural (target) RNA sequence. A higher recovery rate indicates greater accuracy in sequence reconstruction
- Perplexity. Computed as the exponentiated average negative log-likelihood of the native sequence, perplexity reflects the model's uncertainty in predicting the ground-truth sequence. Lower perplexity values indicate better alignment between the model's predictions and the actual sequence.
- Macro-F1 score. The Macro-F1 score is the unweighted average of the F1 scores computed for each nucleotide class (A, U, C and G). It provides a balanced assessment of the model's predictive performance across all nucleotide types, especially useful in the presence of class imbalance.
- TM score. The TM score evaluates the structural similarity between the predicted RNA structure and the native structure. It ranges from 0 to 1, with values closer to 1 indicating higher structural fidelity. This metric emphasizes topological correctness rather than local deviations, making it suitable for assessing global structural accuracy.
- Root mean square deviation (RMSD). This metric quantifies the average distance between corresponding atoms (typically backbone atoms) in the predicted and native structures after optimal superposition. Lower RMSD values indicate higher structural fidelity.
- GDT_TS (Global Distance Test – Total Score). This metric evaluates the percentage of correctly predicted nucleotides within defined distance thresholds (for example, 1 Å, 2 Å, 4 Å and 8 Å) between predicted and native structures. GDT_TS provides a balanced view of both local and global structural accuracy.

For each RNA, we select the C4′ coordinates to measure TM-score, RMSD, and GDT_TS.

- Secondary structure MCC (Matthews correlation coefficient). This metric assesses the agreement between predicted and native secondary structure, using the MCC. The MCC ranges from −1 to 1, with values near 1 indicating strong agreement and values near 0 or below suggesting poor or random prediction.
- Diversity. This metric is measured by computing the average nucleotide difference across all pairwise comparisons within a set of generated sequences. Specifically, it is calculated as the proportion of differing nucleotide positions between two sequences, averaged over all unique sequence pairs in the generated set. A higher diversity score indicates a broader exploration of the sequence space, reflecting the model's ability to generate varied RNA sequences.
- KL divergence. It measures the difference between the model's predicted nucleotide distribution at each position and the empirical distribution derived from the MSA. Lower KL divergence indicates that the model better captures the conservation and variation patterns observed in naturally occurring homologous sequences.
- Absolute entropy difference. It computes the absolute difference in entropy between the model's predicted nucleotide distribution and the MSA-derived distribution at each position. This metric evaluates how well the model reproduces the uncertainty or variability present in the MSA. A smaller absolute entropy difference suggests that the model aligns well with the natural sequence variability.

Statistical analysis

To assess the robustness of our findings, all quantitative metrics (for example, recovery rate, perplexity and foldability) were evaluated across three independent runs, each initialized with a different random seed. The reported values are expressed as mean ± standard deviation (s.d.) across three seeds.

For pairwise method comparisons, one-sided paired Wilcoxon signed-rank tests were conducted to determine whether AlignIF achieves statistically higher performance. For correlation analyses, 95% CIs were estimated using bootstrap resampling with 5,000 replicates. Specifically, sample pairs were resampled with replacement, and correlation coefficients were recomputed for each bootstrap sample. The 2.5th and 97.5th percentiles of the bootstrap distribution were taken as the bounds of the 95% CI.

Wet-lab experiments for AlignIF-designed RNA sequences

RNA preparation. All oligonucleotide sequences used in the study are listed in Supplementary Data 1. DNA template oligonucleotides were synthesized by Sangon Biotech or GENEWIZ. The RNA substrate oligonucleotide of the pistol ribozyme was synthesized by GENEWIZ. All RNAs were transcribed from double-stranded DNA templates containing a 5′ T7 RNA polymerase promoter. In vitro transcription was carried out in a 20-µl reaction containing 400 ng DNA template, 1× transcription buffer (B110085, Sangon Biotech), 7.5 mM ribonucleoside triphosphates (B600057, Sangon Biotech), 1 U μl^{-1} RNase inhibitor (B600478, Sangon Biotech), 5 U ml^{-1} yeast inorganic pyrophosphatase (B110084, Sangon Biotech) and 2.5 U μl^{-1} T7 RNA polymerase (B110085, Sangon Biotech) at 37 °C for 12 h. All RNA transcripts were treated with 0.1 U μl^{-1} DNaseI (B618252, Sangon Biotech) in 1× reaction buffer with $MgCl_2$ (B618252, Sangon Biotech) at 37 °C for 30 min (total volume 40 µl). The reaction was quenched by adding ethylenediaminetetraacetic acid to a final concentration of 5 mM, followed by heat inactivation of DNase I at 65 °C. The reaction mixture was purified using a spin column RNA cleanup & concentration kit (B518688, Sangon Biotech). The RNA was eluted in DEPC-treated H_2O and then quantified using a Nanodrop spectrophotometer (Fisher Scientific).

Fluorescence measurements of generated fluorescent RNA aptamers. In vitro reactions were performed on a black 96-well microplate with each of the generated fluorescent RNA aptamers at a final concentration of 2 µM and TO1-biotin (TO1-3PEG-Biotin, G7955, Applied Biological Materials) at a final concentration of 4 µM in reaction buffer (containing 10 mM NaH_2PO_4, pH 7.2, 140 mM KCl, 1 mM $MgCl_2$ and 0.05% Tween-20). Reactions were incubated at 37 °C for 30 min, and fluorescence was measured at excitation/emission = 500/545 using a Spark multimode microplate reader (TECAN). *i*Mango-III and Mango-I were used as positive controls for normalization. Fluorescence visualization through the Cy2 channel by Typhoon biomolecular imager (Cytiva).

K_d value measurements were performed under the same conditions of fluorescence, but TO1-biotin at a series of final concentrations of 31.25, 62.5, 125, 250, 500, 1,000 and 2,000 nM. Fluorescence of the target aptamer binding with 2,000 nM TO1-biotin, as normalization. K_d values were calculated by fitting one site-specific binding model using Prism 9, GraphPad. Quantification data were obtained from three independent replicates.

CD spectroscopy. Generated aptamers and Mango-I aptamer (2 µM) were prepared in 10 mM NaH_2PO_4 (pH 7.2) buffer containing 4 µM TO1-biotin, 140 mM KCl and 1 mM $MgCl_2$, or containing only 4 µM TO1-biotin. All RNA samples were refolded through a heating/cooling process before the experiments were conducted. CD spectra were recorded at 25 °C and 230–320 nm on a Chirascan V100 (Applied Photophysics) in a 0.5-mm path-length quartz cuvette. Each sample acquisition signal was collected three times, and the corresponding background buffer signal was subtracted from each measurement.

Single-turnover reactions of the generated pistol ribozymes in trans

Kit-purified generated pistol ribozymes and RNA substrates were mixed, annealed and reacted in cleavage buffer (30 mM HEPES pH 7.5, 100 mM KCl and 10 mM $MgCl_2$) at 23 °C. Reactions were stopped by the addition of polyacrylamide gel electrophoresis (PAGE) gel loading buffer (95% formamide, 25 mM ethylenediaminetetraacetic acid pH 7.5) and resolved by 15% urea-PAGE. FAM-labeled RNA substrates were visualized using a Typhoon biomolecular imager (Cytiva) and quantified using ImageQuant TL (Cytiva).

Pseudo first-order reaction rates (k_{obs}) under single-turnover pre-steady-state (K_m/k_{cat}) conditions were determined using 1 µM ribozyme and 50 nM substrate. Samples were taken and reactions stopped at time points (0, 1, 2, 5, 10, 20 and 40 min) on ice in 6× excess PAGE gel loading buffer. The mixtures were denatured at 95 °C for 15 min and then rapidly cooled to 4 °C. Samples were analyzed by 15% urea-PAGE, visualized and quantified using ImageQuant TL. Quantification data from three independent replicates. k_{obs} values were calculated by fitting the percentage of substrate cleaved and reaction time to a one-phase decay model using Prism 9, GraphPad.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in the article and its Supplementary Information. The training data are obtained from the BGSU RNA Hub website (https://rna.bgsu.edu/rna3dhub/). The PDB structures for *i*Mango-III (chain B of 6E8U), Mango-I (chain A of 5V3F), and env25 (chain A of 5K7C) are available from the PDB. Processed datasets, train/validation/test splits and MStA ensemble members used for each target are available via Zenodo at https://doi.org/10.5281/zenodo.20528802 (ref. 57). The designed RNA sequences for wet-lab experiments and all oligonucleotide sequences are provided in Supplementary Data 1. Source data are provided with this paper.

Code availability

Source codes are available via GitHub at https://github.com/sfwang-auto/AlignIF and via Zenodo at https://doi.org/10.5281/zenodo.20531073 (ref. 58). A frozen release in GitHub (tag v1.2, commit hash: 831f857) has been created to ensure reproducibility, including environment specifications, trained model checkpoints and a one-command script to reproduce major results. To facilitate the use of AlignIF, we developed a user-friendly web server at http://www.csbio.sjtu.edu.cn/bioinf/AlignIF/.

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Acknowledgements

We acknowledge the support from the Shared Instrumentation Core Facility, Hangzhou Institute of Medicine (HIM) Chinese Academy of Sciences.

Author contributions

X.P., J.S. and H.-B.S. conceptualized the project. S.W. was responsible for dataset collection and preprocessing. The model was developed by S.W., with methodological guidance provided by X.P. and J.Y., and S.W. carried out model training. X.P., J.S. and H.-B.S. supervised and provided strategic direction for the project. S.W. developed the online website and implemented its functionality. J.W. and J.S. experimentally verified the effectiveness of AlignIF. S.W., J.W., X.L., W.Z., J.H., Y.X. and X.P. drafted the paper, with contributions from J.S. and H.-B.S. All authors reviewed and approved the final paper.

Funding

This work is supported by the National Natural Science Foundation of China (grant no. 62473257), and the Science and Technology Commission of Shanghai Municipality (grant nos. 24ZR1435300 and 22511104100), and the National Key Research and Development Program of China (grant no. 2022YFC3400400), Leading Health Talents in Zhejiang Province (grant no. WS2022LJ01) and the Natural Science Foundation of Zhejiang Province (grant no. YXD23B0301).

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s43588-026-01029-2.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43588-026-01029-2.

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Peer review information *Nature Computational Science* thanks Mohammad Mohebbi and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editors: Fernando Chirigati and Ananya Rastogi, in collaboration with the *Nature Computational Science* team. Peer reviewer reports are available.

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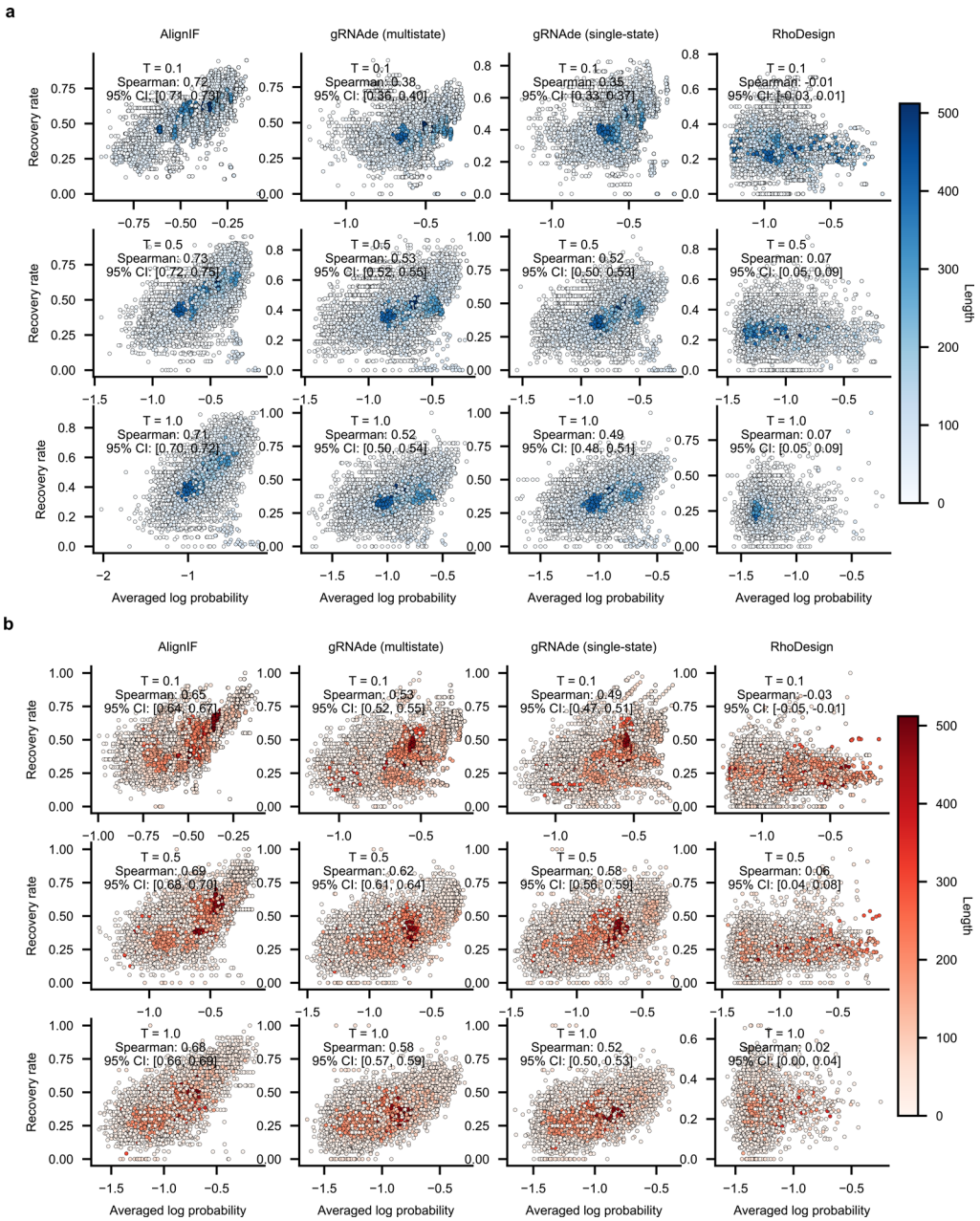

Extended Data Fig. 1 | Correlation between sequence recovery rate and averaged log probability. For each structure in (**a**) the benchmark test set and (**b**) the independent test set, 20 sequences were generated using sampling temperatures of 0.1, 0.5, and 1.0 across three random seeds. Each point represents one generated sequence, with sequence recovery rate plotted against its average log probability. Point color intensity indicates sequence length. Spearman correlation coefficients with 95% CIs between recovery rate and averaged log probability are shown in the upper-right corner of each panel.

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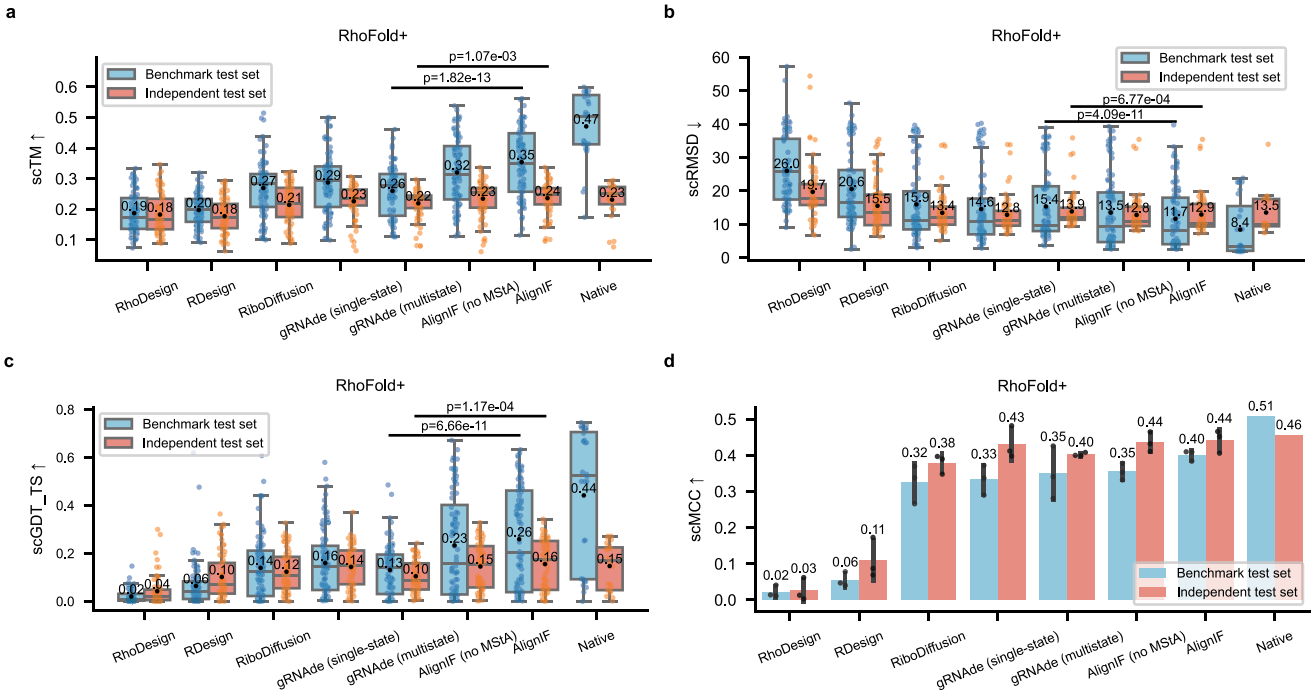

Extended Data Fig. 2 | Foldability evaluation of RNA sequences designed by AlignIF and other methods using RhoFold + . a-c, Three tertiary structure self-consistency metrics: (**a**) scTM score; (**b**) scRMSD; (**c**) scGDT_TS; across multiple methods and three random seeds. Box centers show medians, box limits show IQRs (25th–75th percentiles), whiskers extend to the most extreme values within 1.5×IQR, and black points indicate means. Each box summarizes three seeds, with n = 96 structures for the benchmark test set and n = 72 structures for the independent test set. The 'native' columns (rightmost) represent upper bounds derived from native sequences by RhoFold + . One-sided paired Wilcoxon signed-rank test p-values between AlignIF and gRNAde (multistate) are shown. **d**, Secondary structure self-consistency evaluated using scMCC (mean ± s.d. across three random seeds).

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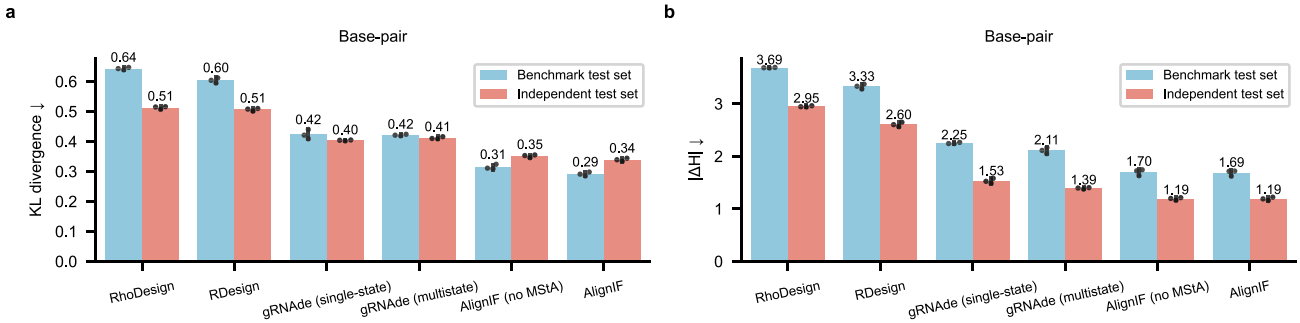

Extended Data Fig. 3 | Evaluation of evolutionary fidelity at base-pair positions for AlignIF and other methods. Evolutionary fidelity is assessed using (**a**) the KL divergence and (**b**) absolute entropy difference (|ΔH |) between model-generated sequence profiles and reference MSA profiles at base-pair positions across three random seeds.

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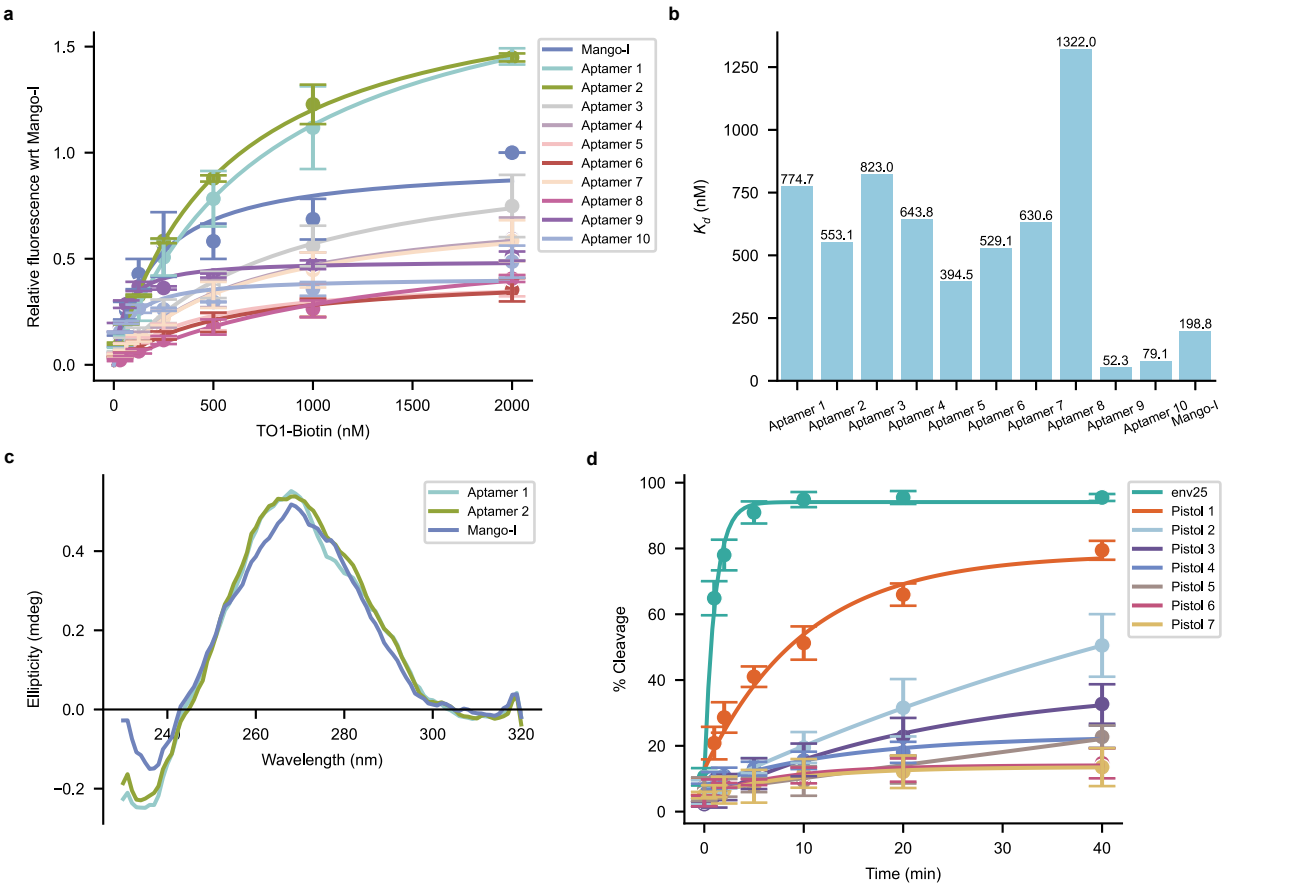

Extended Data Fig. 4 | Mechanism investigation of light-up RNA aptamers and experimental validation of pistol ribozymes generated by AlignIF. **a**, Measurement of K_d for Mango-I and aptamers 1 to 10. 4 μM RNA incubated with a series of final concentrations of 31.25, 62.5, 125, 250, 500, 1000, and 2000 nM TO1-biotin. Fluorescence normalized by Mango-I with 2000 nM TO1-biotin. The normalized fluorescence values were fitted using Prism 9, GraphPad. Error bars show the standard error of the mean of three independent replicates. **b**, K_d values of aptamers 1 to 10 and Mango-I. **c**, CD spectroscopy of aptamers 1, aptamer 2, and Mango-I. 2 μM RNA incubated with 4 μM TO1-biotin in 10 mM NaH_2PO_4 (pH 7.2) buffer. CD spectra were recorded at 25 °C and 230 to 320 nm in a 0.5-mm path-length quartz cuvette, with three independent replicates. **d**, Pre-steady-state bimolecular reaction of 50 nM substrate with 1 μM pistol 1 to 7 and env25 (23 °C, pH 7.5, 10 mM Mg^{2+}; mean ± s.d. across three independent experimental replicates).

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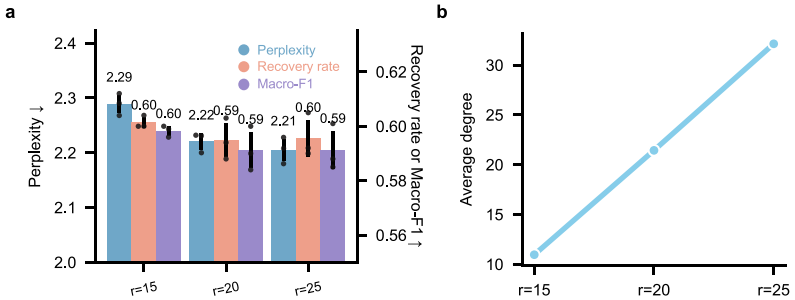

Extended Data Fig. 5 | Sensitivity analysis of model performance to the neighbor cutoff distance. a, Model performance, measured by perplexity, sequence recovery rate, and Macro-F1 score (mean ± s.d. across three random seeds) evaluated at varying neighbor cutoffs. **b**, The average node degree increases approximately linearly with the larger neighbor cutoffs, reflecting the inclusion of more spatial neighbors per nucleotide.

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Software and code

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Data collection	No software was used for data collection.
Data analysis	AligIF is available at https://github.com/sfwang-auto/AlignIF (no version number). US-align is available at https://github.com/pylelab/USalign (no version number). BioPython (v.1.85) is available at https://biopython.org/wiki/Download. RhoDesign is available at https://github.com/ml4bio/RhoDesign (no version number). RDesign is available at https://github.com/A4Bio/RDesign (no version number). RiboDiffusion is available at https://github.com/ml4bio/RiboDiffusion (no version number). gRNAde is available at https://github.com/chaitjo/geometric-rna-design (no version number). AlphaFold3 is available at https://github.com/google-deepmind/alphafold3 (no version number).

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All data are available in the paper and the Supplementary Information. The training data are obtained from the BGSU RNA Hub website (https://rna.bgsu.edu/rna3dhub/). Source data for Figures 2-6 and Extended Data Figures 1-5 are available with this manuscript. The PDB structures for iMango-III (chain B of 6E8U), Mango-I (chain A of 5V3F), and env25 (chain A of 5K7C) are available from the PDB. Processed datasets, train/validation/test splits, and MStA ensemble members used for each target are available in Zenodo at https://zenodo.org/records/20528802 (ref.57). The designed RNA sequences for wet-lab experiments and all oligonucleotide sequences are provided in Supplementary Data 1.

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Life sciences study design

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Sample size	No statistical method was used to predetermine sample size. For computational benchmarking, the sample size was determined by the availability of non-redundant RNA structures in the BGSU RNA 3D Hub, which includes 7,932 RNAs clustered into 499 families. These were partitioned into training, validation, and test sets. An independent test set of 186 RNAs was curated based on temporal and structural dissimilarity to ensure unbiased evaluation. For wet-lab validation, we synthesized 13 iMango-III–derived aptamers, 10 Mango-I–derived aptamers, and 7 pistol ribozyme variants. All fluorescence, binding, spectroscopy, and cleavage assays were performed with three independent biological replicates (n = 3). We determined these sample sizes to be sufficient for the exploratory nature of this study, enabling confirmation of functional activity across all tested sequences (e.g., fluorescence in 23/23 aptamers, cleavage in 7/7 ribozymes), identification of performance improvements (e.g., two aptamers exhibiting >1.4-fold increased brightness), and demonstration of reproducible trends supported by error estimates. Together, these results provide robust proof-of-concept validation without the need for larger cohorts at this stage.
Data exclusions	No data were excluded.
Replication	All fluorescence, binding, spectroscopy, and cleavage assays were performed with three independent replicates. All replication attempts produced consistent results, supporting the reproducibility of the experimental findings.
Randomization	Samples were allocated into experimental groups randomly.
Blinding	Blinding was not applicable in this study, as data collection and analysis were conducted using automated computational methods, and all wet-lab experiments relied on objective, quantitative measurements. Techniques such as fluorescence spectroscopy, binding assays for dissociation constants, circular dichroism spectroscopy, and gel electrophoresis with rate constant calculations for ribozyme cleavage activity were performed using standardized instrumental protocols. These methods yield quantifiable, instrument-derived data that are not subject to

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